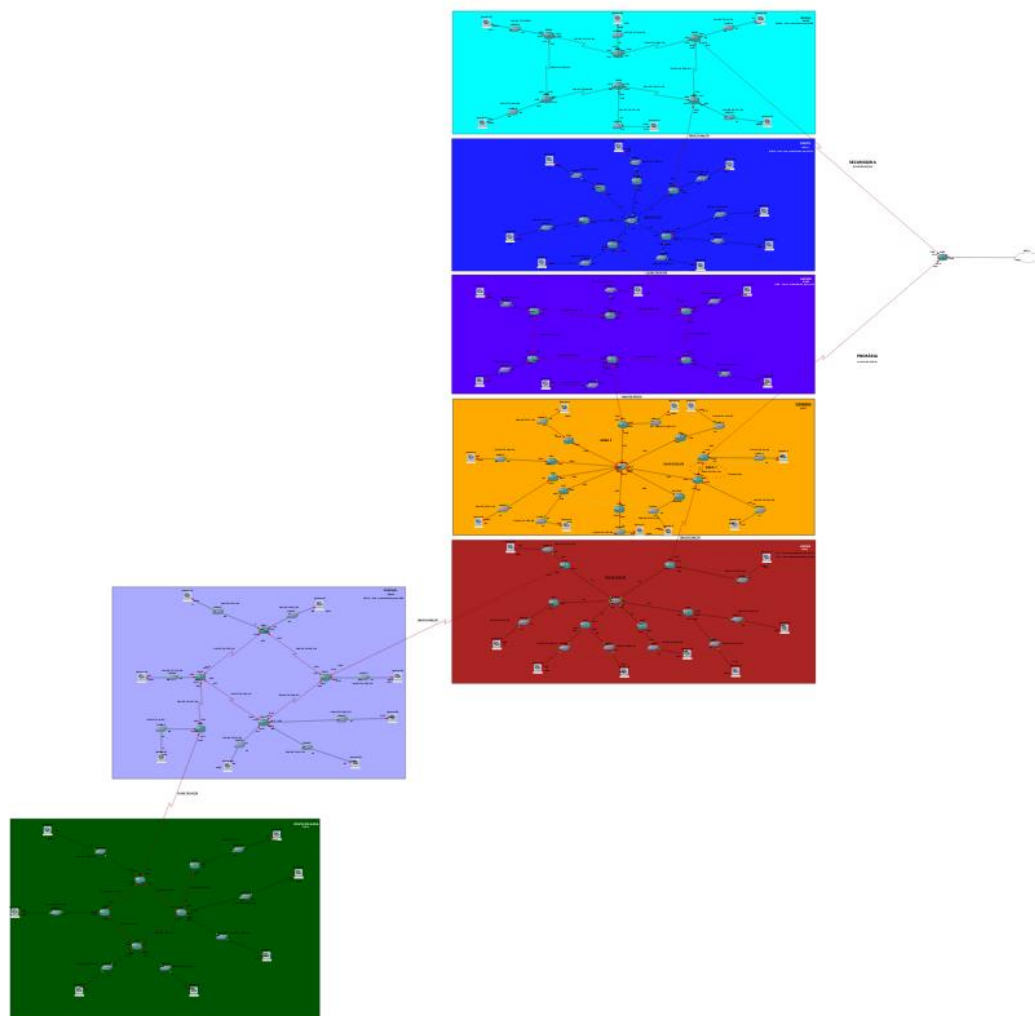


Instituto Superior de Engenharia de Coimbra

Licenciatura em Engenharia Informática

Encaminhamento de Dados



Docente: Amâncio Santos

*Coimbra, 23 de maio
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Nº 2021145193*

- Mapa de Endereçamento:

Subnet	Needed Size	Allocated	Address	Mask	Dec Mask	Assignable Range	Broadcast
LAN1-B	4	6	194.65.74.24	/29	255.255.255.248	194.65.74.25 - 194.65.74.30	194.65.74.31
B1-B2	2	2	194.65.74.184	/30	255.255.255.252	194.65.74.185 - 194.65.74.186	194.65.74.187
B1-B6	2	2	194.65.74.188	/30	255.255.255.252	194.65.74.189 - 194.65.74.190	194.65.74.191
LAN2-B	6	6	194.65.73.168	/29	255.255.255.248	194.65.73.169 - 194.65.73.174	194.65.73.175
B2-B3	2	2	194.65.74.192	/30	255.255.255.252	194.65.74.193 - 194.65.74.194	194.65.74.195
LAN3-B	8	14	194.65.72.240	/28	255.255.255.240	194.65.72.241 - 194.65.72.254	194.65.72.255
B3-B4	2	2	194.65.74.196	/30	255.255.255.252	194.65.74.197 - 194.65.74.198	194.65.74.199
LAN4-B	10	14	194.65.72.160	/28	255.255.255.240	194.65.72.161 - 194.65.72.174	194.65.72.175
B4-B5	2	2	194.65.74.200	/30	255.255.255.252	194.65.74.201 - 194.65.74.202	194.65.74.203
LAN5-B	12	14	194.65.72.112	/28	255.255.255.240	194.65.72.113 - 194.65.72.126	194.65.72.127
B5-B6	2	2	194.65.74.204	/30	255.255.255.252	194.65.74.205 - 194.65.74.206	194.65.74.207
LAN6-B	4	6	194.65.74.112	/29	255.255.255.248	194.65.74.113 - 194.65.74.118	194.65.74.119
LIG_P	2	2	194.65.74.244	/30	255.255.255.252	194.65.74.245 - 194.65.74.246	194.65.74.247
LAN1-P	6	6	194.65.73.152	/29	255.255.255.248	194.65.73.153 - 194.65.73.158	194.65.73.159
STAR-P	6	6	194.65.74.16	/29	255.255.255.248	194.65.74.17 - 194.65.74.22	194.65.74.23
LAN2-P	6	6	194.65.73.184	/29	255.255.255.248	194.65.73.185 - 194.65.73.190	194.65.73.191
LAN3-P	8	14	194.65.73.16	/28	255.255.255.240	194.65.73.17 - 194.65.73.30	194.65.73.31
LAN4-P	8	14	194.65.73.64	/28	255.255.255.240	194.65.73.65 - 194.65.73.78	194.65.73.79
LAN5-P	4	6	194.65.74.96	/29	255.255.255.248	194.65.74.97 - 194.65.74.102	194.65.74.103
LIG_A	2	2	194.65.74.228	/30	255.255.255.252	194.65.74.229 - 194.65.74.230	194.65.74.231
LAN6-P	4	6	194.65.74.128	/29	255.255.255.248	194.65.74.129 - 194.65.74.134	194.65.74.135
LAN7-P	6	6	194.65.74.0	/29	255.255.255.248	194.65.74.1 - 194.65.74.6	194.65.74.7
LAN8-P	4	6	194.65.74.144	/29	255.255.255.248	194.65.74.145 - 194.65.74.150	194.65.74.151
LAN1-A	6	6	194.65.73.128	/29	255.255.255.248	194.65.73.129 - 194.65.73.134	194.65.73.135
A1-A2	2	2	194.65.74.160	/30	255.255.255.252	194.65.74.161 - 194.65.74.162	194.65.74.163
LAN2-A	6	6	194.65.73.160	/29	255.255.255.248	194.65.73.161 - 194.65.73.166	194.65.73.167
A2-A3	2	2	194.65.74.164	/30	255.255.255.252	194.65.74.165 - 194.65.74.166	194.65.74.167
LAN3-A	4	6	194.65.74.64	/29	255.255.255.248	194.65.74.65 - 194.65.74.70	194.65.74.71
A3-A4	2	2	194.65.74.168	/30	255.255.255.252	194.65.74.169 - 194.65.74.170	194.65.74.171
LAN4-A	8	14	194.65.73.32	/28	255.255.255.240	194.65.73.33 - 194.65.73.46	194.65.73.47
A4-A5	2	2	194.65.74.172	/30	255.255.255.252	194.65.74.173 - 194.65.74.174	194.65.74.175
LAN5-A	12	14	194.65.72.96	/28	255.255.255.240	194.65.72.97 - 194.65.72.110	194.65.72.111
A5-A6	2	2	194.65.74.176	/30	255.255.255.252	194.65.74.177 - 194.65.74.178	194.65.74.179
LIG_C	2	2	194.65.74.232	/30	255.255.255.252	194.65.74.233 - 194.65.74.234	194.65.74.235
LAN6-A	4	6	194.65.74.104	/29	255.255.255.248	194.65.74.105 - 194.65.74.110	194.65.74.111
A6-A1	2	2	194.65.74.180	/30	255.255.255.252	194.65.74.181 - 194.65.74.182	194.65.74.183
LAN1-C	4	6	194.65.74.32	/29	255.255.255.248	194.65.74.33 - 194.65.74.38	194.65.74.39

STAR-C	9	14	194.65.72.224	/28	255.255.255.240	194.65.72.225 - 194.65.72.238	194.65.72.239
LAN2-C	6	6	194.65.73.176	/29	255.255.255.248	194.65.73.177 - 194.65.73.182	194.65.73.183
LAN3-C	6	6	194.65.73.200	/29	255.255.255.248	194.65.73.201 - 194.65.73.206	194.65.73.207
LAN4-C	4	6	194.65.74.72	/29	255.255.255.248	194.65.74.73 - 194.65.74.78	194.65.74.79
LAN5-C	6	6	194.65.73.224	/29	255.255.255.248	194.65.73.225 - 194.65.73.230	194.65.73.231
LAN6-C	8	14	194.65.73.80	/28	255.255.255.240	194.65.73.81 - 194.65.73.94	194.65.73.95
LAN7-C	12	14	194.65.72.128	/28	255.255.255.240	194.65.72.129 - 194.65.72.142	194.65.72.143
LAN8-C	8	14	194.65.73.112	/28	255.255.255.240	194.65.73.113 - 194.65.73.126	194.65.73.127
LAN9-C	4	6	194.65.74.152	/29	255.255.255.248	194.65.74.153 - 194.65.74.158	194.65.74.159
LAN10-C	10	14	194.65.72.144	/28	255.255.255.240	194.65.72.145 - 194.65.72.158	194.65.72.159
LIG_L	2	2	194.65.74.240	/30	255.255.255.252	194.65.74.241 - 194.65.74.242	194.65.74.243
C01-C010	2	2	194.65.75.16	/30	255.255.255.252	194.65.75.17 - 194.65.75.18	194.65.75.19
LAN1-L	6	6	194.65.73.144	/29	255.255.255.248	194.65.73.145 - 194.65.73.150	194.65.73.151
LIG_F	2	2	194.65.74.236	/30	255.255.255.252	194.65.74.237 - 194.65.74.238	194.65.74.239
STAR-L	6	6	194.65.74.8	/29	255.255.255.248	194.65.74.9 - 194.65.74.14	194.65.74.15
LAN2-L	4	6	194.65.74.56	/29	255.255.255.248	194.65.74.57 - 194.65.74.62	194.65.74.63
LAN3-L	6	6	194.65.73.208	/29	255.255.255.248	194.65.73.209 - 194.65.73.214	194.65.73.215
LAN4-L	8	14	194.65.73.48	/28	255.255.255.240	194.65.73.49 - 194.65.73.62	194.65.73.63
LAN5-L	10	14	194.65.72.192	/28	255.255.255.240	194.65.72.193 - 194.65.72.206	194.65.72.207
LAN6-L	4	6	194.65.74.120	/29	255.255.255.248	194.65.74.121 - 194.65.74.126	194.65.74.127
LAN7-L	6	6	194.65.73.248	/29	255.255.255.248	194.65.73.249 - 194.65.73.254	194.65.73.255
LAN8-L	10	14	194.65.72.208	/28	255.255.255.240	194.65.72.209 - 194.65.72.222	194.65.72.223
LAN1-F	6	6	194.65.73.136	/29	255.255.255.248	194.65.73.137 - 194.65.73.142	194.65.73.143
F1-F2	2	2	194.65.74.208	/30	255.255.255.252	194.65.74.209 - 194.65.74.210	194.65.74.211
LAN2-F	4	6	194.65.74.48	/29	255.255.255.248	194.65.74.49 - 194.65.74.54	194.65.74.55
LAN3-F	8	14	194.65.73.0	/28	255.255.255.240	194.65.73.1 - 194.65.73.14	194.65.73.15
F2-F3	2	2	194.65.74.212	/30	255.255.255.252	194.65.74.213 - 194.65.74.214	194.65.74.215
LAN4-F	10	14	194.65.72.176	/28	255.255.255.240	194.65.72.177 - 194.65.72.190	194.65.72.191
F3-F4	2	2	194.65.74.216	/30	255.255.255.252	194.65.74.217 - 194.65.74.218	194.65.74.219
LAN5-F	4	6	194.65.74.88	/29	255.255.255.248	194.65.74.89 - 194.65.74.94	194.65.74.95
LAN6-F	8	14	194.65.73.96	/28	255.255.255.240	194.65.73.97 - 194.65.73.110	194.65.73.111
LAN7-F	6	6	194.65.73.240	/29	255.255.255.248	194.65.73.241 - 194.65.73.246	194.65.73.247
F4-F1	2	2	194.65.74.220	/30	255.255.255.252	194.65.74.221 - 194.65.74.222	194.65.74.223
LAN8-F	32	62	194.65.72.0	/26	255.255.255.192	194.65.72.1 - 194.65.72.62	194.65.72.63
F5-F3	2	2	194.65.74.224	/30	255.255.255.252	194.65.74.225 - 194.65.74.226	194.65.74.227
LIG_PD	2	2	194.65.74.248	/30	255.255.255.252	194.65.74.249 - 194.65.74.250	194.65.74.251
LAN1-PD	4	6	194.65.74.40	/29	255.255.255.248	194.65.74.41 - 194.65.74.46	194.65.74.47
PD1-PD2	2	2	194.65.74.252	/30	255.255.255.252	194.65.74.253 - 194.65.74.254	194.65.74.255
LAN2-PD	6	6	194.65.73.192	/29	255.255.255.248	194.65.73.193 - 194.65.73.198	194.65.73.199
PD2-PD3	2	2	194.65.75.0	/30	255.255.255.252	194.65.75.1 - 194.65.75.2	194.65.75.3
LAN3-PD	6	6	194.65.73.216	/29	255.255.255.248	194.65.73.217 - 194.65.73.222	194.65.73.223
LAN4-PD	4	6	194.65.74.80	/29	255.255.255.248	194.65.74.81 - 194.65.74.86	194.65.74.87
PD3-PD4	2	2	194.65.75.4	/30	255.255.255.252	194.65.75.5 - 194.65.75.6	194.65.75.7
LAN5-PD	6	6	194.65.73.232	/29	255.255.255.248	194.65.73.233 - 194.65.73.238	194.65.73.239
LAN6-PD	4	6	194.65.74.136	/29	255.255.255.248	194.65.74.137 - 194.65.74.142	194.65.74.143
PD4-PD1	2	2	194.65.75.8	/30	255.255.255.252	194.65.75.9 - 194.65.75.10	194.65.75.11
LAN7-PD	16	30	194.65.72.64	/27	255.255.255.224	194.65.72.65 - 194.65.72.94	194.65.72.95
PD5-PD4	2	2	194.65.75.12	/30	255.255.255.252	194.65.75.13 - 194.65.75.14	194.65.75.15

1. Introdução

Este trabalho tem como finalidade o planeamento de uma rede de dados local e alargada de uma empresa fictícia de acordo a desafiar as competências do aluno no que toca ao planeamento do projeto com base na configuração de routers no sistema operativo Cisco IOS/IOU.

Nas imagens acima é possível encontrar o mapa de endereçamento do projeto, e na capa, a respetiva topologia.

2. Endereçamento

O endereçamento atribuído foi um 194.65.72.0/21, sendo o “72” proveniente do cálculo matemático: $8 \times (2021145193 \bmod 32)$.

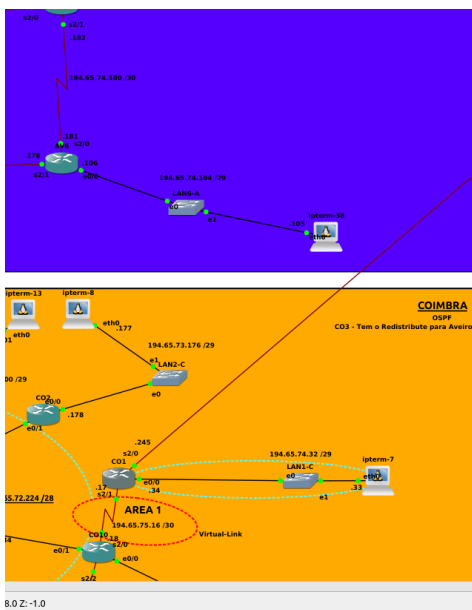
Devido a um equívoco da minha parte, não foi possível a utilização de endereçamento privado.

3. Filiais

Todos os protocolos pedidos no enunciado foram usados nas filiais com a devida autenticação, assim como também foram mudadas as larguras de banda de cada link. Todos os routers contém uma única autenticação por telnet e é apresentado um banner aquando da entrada no mesmo.

4. Ligação ao Exterior

De modo a ter ligação ao exterior, foi configurada a *default route* em Coimbra e em Braga, de modo que, com a saída primária ativa ou não, tenhamos uma segunda opção de conexão ao exterior. Nas seguintes imagens, poderemos observar na primeira a rota com a saída primária ativa e na segunda com apenas a saída secundária ativa, por Coimbra e Braga, respetivamente.



PRIMÁRIA
10.200.93.246/30

```

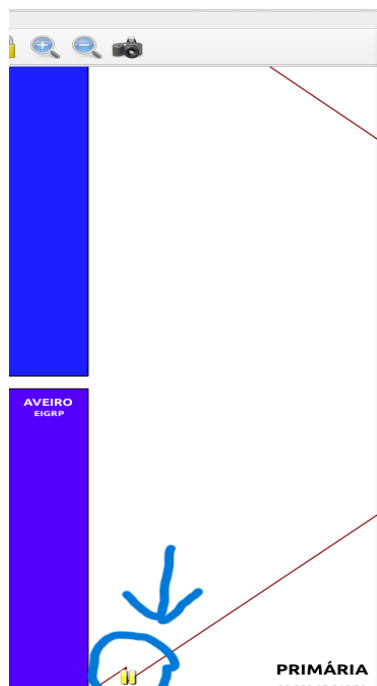
CO1#
password 7 110A1016141D
login
transport input telnet
line vty 1 4
login
transport input none
!
!
end

CO1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

S*    0.0.0.0/0 is directly connected, Serial2/0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      C    10.200.93.244/30 is directly connected, Serial2/0
      L    10.200.93.245/32 is directly connected, Serial2/0
      L    194.65.72.0/24 is variably subnetted, 10 subnets, 4 masks
      O E2  194.65.72.0/24 [110/20] via 194.65.75.18, 00:00:19, Serial2/1
      O E2  194.65.72.0/26 [110/20] via 194.65.75.18, 00:01:05, Serial2/1
      O E2  194.65.72.64/27 [110/28] via 194.65.75.18, 00:00:33, Serial2/1
      O E2  194.65.72.96/28 [110/28] via 194.65.75.18, 00:00:19, Serial2/1
      O IA  194.65.72.128/28 [110/84] via 194.65.75.18, 00:00:24, Serial2/1

CO1#
  
```



PRIMÁRIA
10.200.93.246/30

```

BRA1#
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

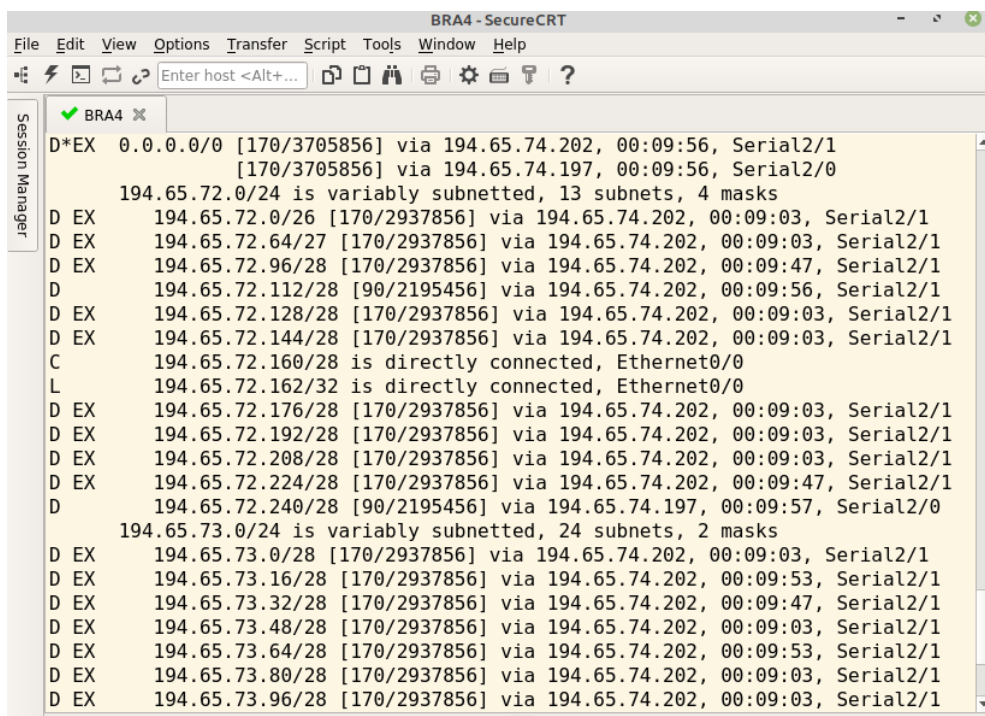
S*    0.0.0.0/0 is directly connected, Serial2/2
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
      C    10.100.93.248/30 is directly connected, Serial2/2
      L    10.100.93.249/32 is directly connected, Serial2/2
      L    194.65.72.0/24 is variably subnetted, 12 subnets, 3 masks
      D EX  194.65.72.0/26 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.72.64/27 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.72.96/28 [170/2425856] via 194.65.74.190, 00:02:50, Serial2/0
      D    194.65.72.112/28 [90/2707456] via 194.65.74.190, 00:03:00, Serial2/0
      D EX  194.65.72.128/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.72.144/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D    194.65.72.160/28 [90/3219456] via 194.65.74.190, 00:03:00, Serial2/0
      D    194.65.72.160/28 [90/3219456] via 194.65.74.186, 00:03:00, Serial2/1
      D EX  194.65.72.176/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.72.192/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.72.208/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.72.224/28 [170/2425856] via 194.65.74.190, 00:02:50, Serial2/0
      D    194.65.72.240/28 [90/2707456] via 194.65.74.186, 00:03:01, Serial2/1
      L    194.65.73.0/24 is variably subnetted, 24 subnets, 2 masks
      D EX  194.65.73.0/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.73.16/28 [170/2425856] via 194.65.74.190, 00:02:57, Serial2/0
      D EX  194.65.73.32/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.73.48/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.73.64/28 [170/2425856] via 194.65.74.190, 00:02:57, Serial2/0
      D EX  194.65.73.80/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.73.96/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.73.112/28 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0
      D EX  194.65.73.128/29 [170/2425856] via 194.65.74.190, 00:02:53, Serial2/0
      D EX  194.65.73.136/29 [170/2425856] via 194.65.74.190, 00:02:06, Serial2/0

BRA1#
  
```

5. Routing

Nas próximas imagens, estão presentes as tabelas de *routing* de um router de cada filial sendo a ordem destas:

- Braga
- Porto
- Aveiro
- Coimbra
- Lisboa
- Funchal
- Ponta Delgada



```
BRA4 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
Enter host <Alt+...
BRA4 x
Session Manager
D*EX 0.0.0.0/0 [170/3705856] via 194.65.74.202, 00:09:56, Serial2/1
      [170/3705856] via 194.65.74.197, 00:09:56, Serial2/0
      194.65.72.0/24 is variably subnetted, 13 subnets, 4 masks
D EX   194.65.72.0/26 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.72.64/27 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.72.96/28 [170/2937856] via 194.65.74.202, 00:09:47, Serial2/1
D      194.65.72.112/28 [90/2195456] via 194.65.74.202, 00:09:56, Serial2/1
D EX   194.65.72.128/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.72.144/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
C      194.65.72.160/28 is directly connected, Ethernet0/0
L      194.65.72.162/32 is directly connected, Ethernet0/0
D EX   194.65.72.176/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.72.192/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.72.208/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.72.224/28 [170/2937856] via 194.65.74.202, 00:09:47, Serial2/1
D      194.65.72.240/28 [90/2195456] via 194.65.74.197, 00:09:57, Serial2/0
      194.65.73.0/24 is variably subnetted, 24 subnets, 2 masks
D EX   194.65.73.0/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.73.16/28 [170/2937856] via 194.65.74.202, 00:09:53, Serial2/1
D EX   194.65.73.32/28 [170/2937856] via 194.65.74.202, 00:09:47, Serial2/1
D EX   194.65.73.48/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.73.64/28 [170/2937856] via 194.65.74.202, 00:09:53, Serial2/1
D EX   194.65.73.80/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
D EX   194.65.73.96/28 [170/2937856] via 194.65.74.202, 00:09:03, Serial2/1
```

Router: **Braga**


```

P0R3 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
Enter host <Alt+...
Session Manager
[+] BRA4 [x] [+] P0R3 [x]
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is 194.65.74.17 to network 0.0.0.0

R* 0.0.0.0/0 [120/2] via 194.65.74.17, 00:00:10, Ethernet0/1
    194.65.72.0/24 is variably subnetted, 10 subnets, 4 masks
R   194.65.72.0/24 [120/2] via 194.65.74.17, 00:00:10, Ethernet0/1
R   194.65.72.0/26 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.64/27 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.96/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.128/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.144/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.176/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.192/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.208/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.72.224/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
    194.65.73.0/24 is variably subnetted, 25 subnets, 4 masks
R   194.65.73.0/24 [120/2] via 194.65.74.17, 00:00:10, Ethernet0/1
R   194.65.73.0/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
C   194.65.73.16/28 is directly connected, Ethernet0/0
L   194.65.73.18/32 is directly connected, Ethernet0/0
R   194.65.73.32/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.48/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.64/28 [120/1] via 194.65.74.20, 00:00:13, Ethernet0/1
R   194.65.73.80/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.96/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.112/28 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.128/29 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.136/29 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.144/29 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.152/29 [120/1] via 194.65.74.17, 00:00:10, Ethernet0/1
R   194.65.73.160/29 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1
R   194.65.73.176/29 [120/1] via 194.65.74.21, 00:00:01, Ethernet0/1

```

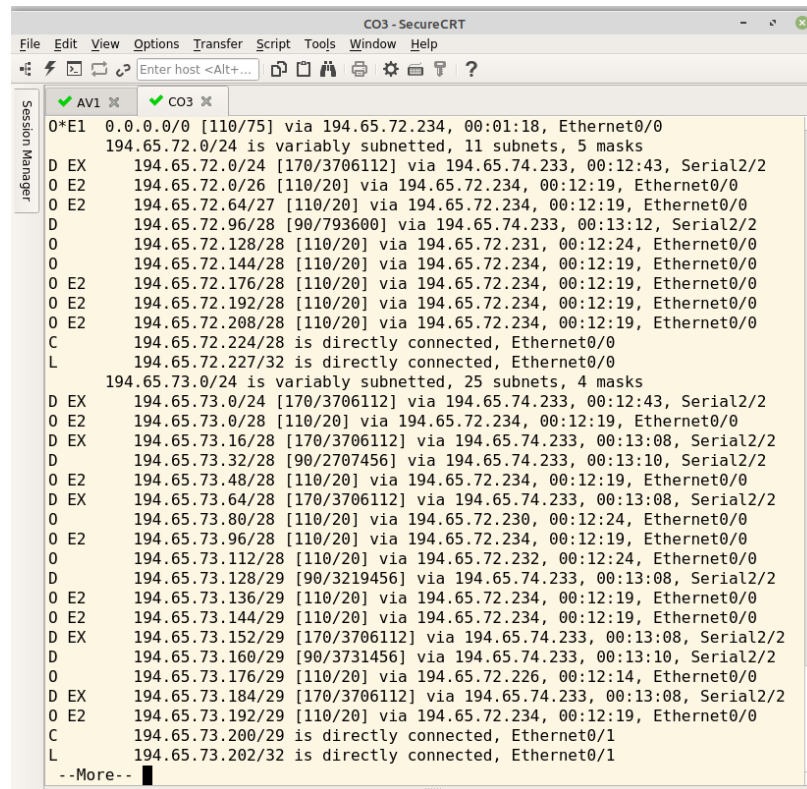
Router: Porto

```

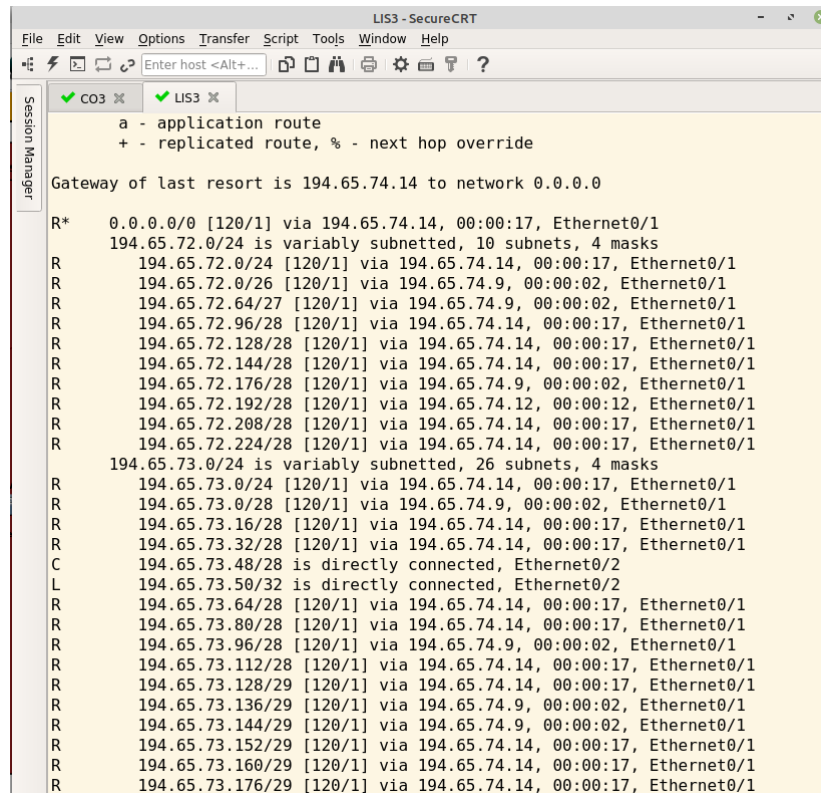
AV1 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
Enter host <Alt+...
Session Manager
[+] AV1 [x]
D*EX 0.0.0.0/0 [170/768256] via 194.65.74.229, 00:12:04, Serial2/2
    194.65.72.0/24 is variably subnetted, 10 subnets, 4 masks
D EX 194.65.72.0/24 [170/768256] via 194.65.74.229, 00:12:04, Serial2/2
D EX 194.65.72.0/26 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.72.64/27 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D   194.65.72.96/28 [90/2707456] via 194.65.74.181, 00:12:29, Serial2/1
D EX 194.65.72.128/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.72.144/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.72.176/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.72.192/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.72.208/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.72.224/28 [170/3449856] via 194.65.74.181, 00:12:29, Serial2/1
    194.65.73.0/24 is variably subnetted, 25 subnets, 4 masks
D EX 194.65.73.0/24 [170/768256] via 194.65.74.229, 00:12:04, Serial2/2
D EX 194.65.73.0/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.16/28 [170/768256] via 194.65.74.229, 00:12:32, Serial2/2
D   194.65.73.32/28 [90/3219456] via 194.65.74.181, 00:12:29, Serial2/1
    [90/3219456] via 194.65.74.162, 00:12:29, Serial2/0
D EX 194.65.73.48/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.64/28 [170/768256] via 194.65.74.229, 00:12:32, Serial2/2
D EX 194.65.73.80/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.96/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.112/28 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
C   194.65.73.128/29 is directly connected, Ethernet0/0
L   194.65.73.130/32 is directly connected, Ethernet0/0
D EX 194.65.73.136/29 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.144/29 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.152/29 [170/768256] via 194.65.74.229, 00:12:32, Serial2/2
D   194.65.73.160/29 [90/2195456] via 194.65.74.162, 00:12:30, Serial2/0
D EX 194.65.73.176/29 [170/3449856] via 194.65.74.181, 00:11:35, Serial2/1
D EX 194.65.73.184/29 [170/768256] via 194.65.74.229, 00:12:32, Serial2/2
D EX 194.65.73.192/29 [170/3449856] via 194.65.74.181, 00:11:45, Serial2/1
D EX 194.65.73.200/29 [170/3449856] via 194.65.74.181, 00:12:29, Serial2/1
--More--

```

Router: Aveiro



Router: Coimbra



Router: Lisboa


```

FNC4 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
Enter host <Alt+...

Session Manager
  LIS3 x FNC4 x
    a - application route
    + - replicated route, % - next hop override

Gateway of last resort is 194.65.74.222 to network 0.0.0.0

D*EX 0.0.0.0/0 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
      194.65.72.0/24 is variably subnetted, 10 subnets, 4 masks
D EX 194.65.72.0/24 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D    194.65.72.0/26 [90/2707456] via 194.65.74.217, 00:13:53, Serial2/1
D EX 194.65.72.64/27 [170/2937856] via 194.65.74.217, 00:13:20, Serial2/1
D EX 194.65.72.96/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.72.128/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.72.144/28 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
D    194.65.72.176/28 [90/2195456] via 194.65.74.217, 00:13:52, Serial2/1
D EX 194.65.72.192/28 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
D EX 194.65.72.208/28 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
D EX 194.65.72.224/28 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
      194.65.73.0/24 is variably subnetted, 26 subnets, 4 masks
D EX 194.65.73.0/24 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D    194.65.73.0/28 [90/2707456] via 194.65.74.222, 00:13:52, Serial2/0
      [90/2707456] via 194.65.74.217, 00:13:52, Serial2/1
D EX 194.65.73.16/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.73.32/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.73.48/28 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
D EX 194.65.73.64/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.73.80/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
C    194.65.73.96/28 is directly connected, Ethernet0/1
L    194.65.73.98/32 is directly connected, Ethernet0/1
D EX 194.65.73.112/28 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.73.128/29 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D    194.65.73.136/29 [90/2195456] via 194.65.74.222, 00:13:52, Serial2/0
D EX 194.65.73.144/29 [170/2937856] via 194.65.74.222, 00:13:52, Serial2/0
D EX 194.65.73.152/29 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0
D EX 194.65.73.160/29 [170/2937856] via 194.65.74.222, 00:13:07, Serial2/0

```

Router: Funchal

```

PD3 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
Enter host <Alt+...

Session Manager
  FNC4 x PD3 x
    Gateway of last resort is 194.65.75.6 to network 0.0.0.0

R* 0.0.0.0/0 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
      194.65.72.0/24 is variably subnetted, 2 subnets, 2 masks
R    194.65.72.0/24 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.72.64/27 [120/2] via 194.65.75.6, 00:00:18, Serial2/1
      194.65.73.0/24 is variably subnetted, 5 subnets, 3 masks
R    194.65.73.0/24 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.73.192/29 [120/1] via 194.65.75.1, 00:00:03, Serial2/0
C    194.65.73.216/29 is directly connected, Ethernet0/0
L    194.65.73.218/32 is directly connected, Ethernet0/0
R    194.65.73.232/29 [120/1] via 194.65.75.6, 00:00:18, Serial2/1
      194.65.74.0/24 is variably subnetted, 44 subnets, 3 masks
R    194.65.74.0/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.8/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.16/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.24/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.40/29 [120/2] via 194.65.75.6, 00:00:18, Serial2/1
      [120/2] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.48/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.56/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.64/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1
      [120/3] via 194.65.75.1, 00:00:03, Serial2/0
R    194.65.74.72/29 [120/3] via 194.65.75.6, 00:00:18, Serial2/1

```

Router: Ponta Delgada

6. Conclusão

Com este trabalho obtive competências práticas e teóricas no que toca a trabalhar em *Cisco*, especialmente na execução de diversos protocolos de encaminhamento, que me fez perceber o funcionamento dos mesmos e como eles comunicam entre si.